

Project 2: Kilns

STAT440/17

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Kilns

According to [United News of Bangladesh](#), the *top three* sources of pollution in Bangladesh in 2021 were cars, brick kilns, and construction. The Bangladeshi government has required that traditional brick kilns be replaced with modern kilns that do not pollute so much. ... Finding the locations of brick kilns in Bangladesh is required to reduce pollution and follow government policy. Air pollution is a leading cause of death in Bangladesh (200,000 deaths in 2025)

- There are two dominant types of kiln technology: *zigzag* (which do not pollute so much) and *fixed chimney kiln*

Dataset

You are given a training dataset consisting of **1,617** labelled satellite photos of kilns (in *train/*.png*)

- The labels are 1 (zigzag kiln) and 0 (fixed chimney kiln) in the file *train/data.csv*
- The images are 256x256 png files with 3 colour channels

You are given a testing dataset consisting of **724** photos (in *test/*.png*). You must predict the labels for these photos. Your predictions must be a score (higher scores mean your model is more confident that the photo is of a zigzag kiln). Your predictions will be evaluated using AUC



Example zigzag kiln

Project 2 logistics

- Rules and evaluation (technical report and contest) will be the same as previous project
- Same teams as for Project 1
- Deadline: November 28th 5pm